

Class Test 4

July 2025

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

L-3/T-2 CSE 321: Computer Networks

Time: 25 minutes

Marks: 20

Student Name: _____

Student No: _____

1. Professor Dumbledore wants to install foolproof IoT security system consisting of 100 sensors in his office-room. He is not that knowledgeable about these muggle technologies and confused what communication protocol he should use for his office-room security system. Notably, the sensors need to just communicate with a server inside the room. Also, they run on batteries, so power consumption is a concern.

His 4 brilliant students suggested 4 different things. Evaluate their comments:

(4 x 4 = 16)

- a) Ron says, "Just connect the sensors with **wires**, easy!"
- b) Harry says, "You should use **Wi-Fi**, it's very fast."
- c) Hermione says, "Use **Bluetooth**, of course! Haven't you read Modern Muggle Technologies?"
- d) Luna says, "I think you should use **LoRaWAN**. It can help you communicate beyond the skies!"

2. One of the motion detecting sensors in Dumbledore's office-room is using MQTT protocol in its application layer. It publishes its data in the topic **"hogwarts/office/motion"**. Ron, Harry and Hermione want to monitor it. Who among them will get the data correctly, and who will not? (2)

- Ron subscribed to **"hogwarts/office/sensors"**
- Hermione subscribed to **"hogwarts/office/motion"**
- Harry subscribed to **"hogwarts/office/#"** (this means all topics under "hogwarts/office")